14 The number of fish in a lake decreases by x% each year.

Given that the number of fish halves in 8 years, work out the value of x.
Give your answer correct to 1 decimal place.

Let r be the multiplier for one year.

$$r^8 = 0.5$$

1 mark
Eight equal multipliers

$r = \sqrt[8]{(0.5)} = 0.9170040432...$

1 mark
Take eighth root

$x = (1 - r) \times 100 = 8.29959568...$

$x = 8.3$ (to 1 decimal place).

1 mark
8.3% per year

(Total for Question 14 is 3 marks)